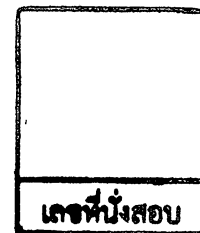


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2017

Subject: 392151 Chemistry I

Section: 15-18

Date: 4 October 2017

Time: 08.00-10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. The examination has 8 pages (including this page), 2 sections and a total score of 60 points.
2. Write all your solution and answer on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are **NOT** allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.

Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.



WebElements: the periodic table on the world-wide web

<http://www.webelements.com/>

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
hydrogen 1 H 1.0079																	helium 2 He 4.0026
lithium 3 Li 6.941	beryllium 4 Be 9.0122											boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180
sodium 11 Na 22.990	magnesium 12 Mg 24.305											aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948
potassium 19 K 39.098	calcium 20 Ca 40.078											gallium 31 Ga 69.723	germanium 32 Ge 72.61	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80
rubidium 37 Rb 85.468	strontium 38 Sr 87.62											indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29
caesium 55 Cs 132.91	barium 56 Ba 137.33	57-70 *										thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]
francium 87 Fr [223]	radium 88 Ra [226]	89-102 **															
			lanthanum 57 La [223]	cerium 58 Ce [226]	praseodymium 59 Pr [229]	neodymium 60 Nd [232]	promethium 61 Pm [234]	samarium 62 Sm [235]	europium 63 Eu [237]	gadolinium 64 Gd [240]	terbium 65 Tb [244]	dysprosium 66 Dy [262]	holmium 67 Ho [263]	erbium 68 Er [267]	thulium 69 Tm [269]	ytterbium 70 Yb [270]	
			actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]	

*lanthanoids

**actinoids

Symbols and names: the symbols and names of the elements, and their spellings are those recommended by the International Union of Pure and Applied Chemistry (IUPAC - <http://www.iupac.org/>). Names have yet to be proposed for the most recently discovered elements 110-112 and 114 so those used here are IUPAC's temporary systematic names. In the USA and some other countries, the spellings aluminum and cesium are normal while in the UK and elsewhere the common spelling is sulphur.

Group labels: the numeric system (1-18) used here is the current IUPAC convention.

Atomic weights (mean relative masses): Apart from the heaviest elements, these are the IUPAC 2001 values and given to 5 significant figures. Elements for which the atomic weight is given within square brackets have no stable nuclides and are represented by the element's longest lived isotope.

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Given :

Speed of light; $c = 3.0 \times 10^8$ m/s

Plank's constant; $h = 6.6 \times 10^{-34}$ J.s

	Multiple Choice	Part 2.1	Part 2.2	Part 2.3	Total
score					

Part 1. Multiple Choice (20 points)

Select the letter that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Question no.	a.	b.	c.	d.
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

- Which of the following is a way in which elements and compounds are similar?
 - Elements and compounds are both pure substances.
 - Elements and compounds are both listed on the periodic table.
 - Elements and compounds are both made up of different kinds of atoms.
 - Elements and compounds can both be broken down by physical changes.
- A water molecule is made up of one oxygen and two hydrogen atoms. Why is water is considered a pure substance?
 - Water can be broken down by physical means.
 - Water can be combined with other substances by physical means.
 - Each water molecule is identical.
 - Water molecules are made up of different types of atoms.

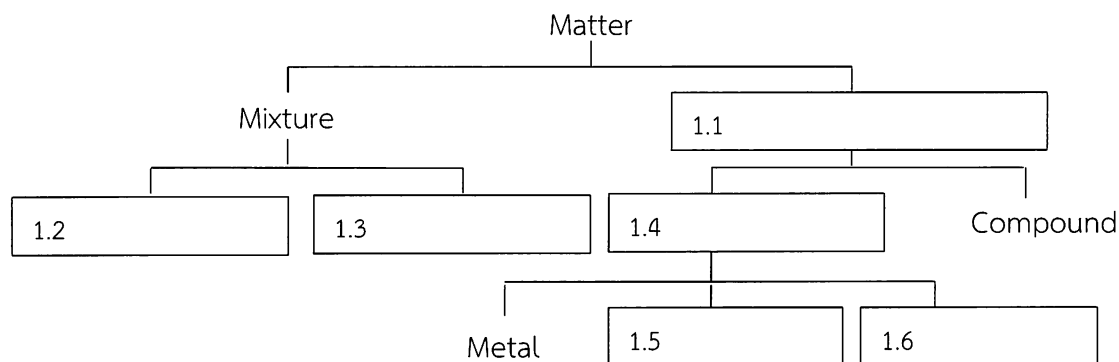
3. If James wants to get the iron filings out of some sand, what could he use to separate these two solids?
- a. A magnet b. Evaporation c. Decanting d. Dialysis
4. Which of these is a solution?
- a. 14K Gold b. Salt Water c. Air d. All of the above
5. Which of the following sentences is incorrect?
- a. Thermal conductivity is the property of a material to conduct heat.
- b. Element can be separated by chemical reaction
- c. Boiling point is an intensive property
- d. Filtration can be used to separate sand from water
6. Which of the following is a physical change?
- a. liquid water evaporates to form water vapor.
- b. The synthesis of water from oxygen and hydrogen.
- c. The decomposition of water to form oxygen and hydrogen.
- d. Heating of iron and sulfur.
7. Which of the following statement is a chemical change?
- a. Separating of sand and rock.
- b. Dissolving of NaCl in water.
- c. Fireworks exploding.
- d. Melting of snow.
8. Which of the following is correct?
- a. Most of synthesis reactions are endothermic reaction.
- b. A product of synthesis reaction is usually larger than the reactants.
- c. Cooling is favored for decomposition reaction.
- d. None of these is correct.
9. Which of the following is an exothermic reaction?
- a. $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
- b. $\text{Fe} + \text{S} \rightarrow \text{FeS}$
- c. $2\text{MgO} \rightarrow 2\text{Mg} + \text{O}_2$
- d. a. and c. are correct.

10. From Rutherford's model of the atom, which of the following statements is not true about the nucleus of this model?
- It contains most of the volume of the atom
 - It is location of neutrons
 - It is positively charged
 - It is the location of the protons
11. Which of following are Isotopes of ${}_{15}^{31}\text{X}$ _____.
- ${}_{15}^{32}\text{X}$
 - ${}_{16}^{31}\text{Y}$
 - ${}_{14}^{30}\text{Z}$
 - none of these is correct
12. The ion of element has 3 positive charge. Mass number of the atom is 27 and the number of neutrons is 14. What is the number of electrons in the ion?
- 13
 - 10
 - 14
 - 16
13. If electrons in an atom have the lowest possible energies, the electrons are in their _____.
- ground states
 - excited states
 - inert states
 - radiation-emitting states
14. Bohr's atomic model _____
- proposes that electrons occupy specific energy levels
 - explains the emission spectra of hydrogen atoms
 - predicts the energy levels of multi-electron atoms
 - both a and b
15. The $n = 1$ shell contains _____ p sub-orbitals. All the other shells contain _____ p sub- orbitals.
- 3, 6
 - 0, 3
 - 6, 2
 - 3, 3
16. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$?
- potassium
 - nitrogen
 - iron
 - vanadium
- 17 The _____ orbital is partially filled in the Manganese atom?
- 3s
 - 4p
 - 4s
 - 3d
18. What is the maximum number of electrons that can occupy the second principle energy level?
- 2
 - 8
 - 10
 - 18
19. All orbitals can hold a maximum of _____ electrons?
- 1
 - 2
 - 3
 - 4
20. Metals like to _____ electrons to become _____ ions.
- lose, positive
 - lose, negative
 - gain, positive
 - gain, negative

Part 2 Write your answer in the space provided for each question (40 points)

1. Complete the diagram for the classification of matter.

(3 points)



2. Classify each of the following as a homogeneous or heterogeneous mixture.

(3 points)

2.1 green tea _____

2.2 tap water _____

2.3 air _____

2.4 salt solution _____

2.5 ice tea _____

2.6 green curry _____

3. Give the name of each of the following substances

(2 points)

3.1 HCl _____

3.2 NO₂ _____

3.3 K₂SO₄ _____

3.4 NH₄OH _____

4. Give the chemical formula for each of the following compounds.

(2 points)

4.1 calcium carbonate _____

4.2 sodium iodide _____

4.3 potassium nitrate _____

4.4 carbon monoxide _____

5. Identify the following as a chemical (C) or physical (P) changes

(2 points)

_____ 5.1 Sugar dissolves in water.

_____ 5.2 Food is digested.

_____ 5.3 Baking soda reacts to vinegar.

_____ 5.4 Alcohol evaporates.

6. For each of the following reactions, say whether it is a synthesis or decomposition reaction and also say whether it is an endothermic or an exothermic reaction:

(2 points)

synthesis/ decomposition

endothermic/exothermic

6.1 $\text{Na}_2\text{CO}_3 \rightarrow 2\text{Na} + \text{O}_2 + \text{CO}_2$ _____

6.2 $4\text{Ag} + \text{O}_2 \rightarrow 2\text{Ag}_2\text{O}$ _____

7. Circle the event in each group that came first (2 points)

7.1 (a) discovery of the nucleus or (b) discovery of the neutron or (c) discovery of the electron

7.2 (a) gold foil experiment or (b) cathode ray experiment or (c) spectrum of hydrogen experiment

8. Draw a Dalton, Thomson and Rutherford model of the neutral carbon-12 (^{12}C) considering the number of subatomic particles). (6 points)

Dalton	Thomson	Rutherford

9. What is the frequency and energy of x-radiation with a 1×10^{-6} m wavelength? (4 points)

10. The following electron configurations belong to element in alkali metal, transition metal, or noble gas? (3 points)

10.1 $1s^2 2s^2 2p^6 3s^2 3p^6$ _____

10.2 $[\text{Kr}]5s^2 4d^{10}$ _____

10.3 2,8,8,1 _____

11. Complete the table below

(4 points)

Element	Atomic number (Z)	Atomic mass (A)	Number of Protons	Number of neutrons	Number of electrons	Number of valence electrons
Nitrogen-14						
Nitrogen-15						
Calcium-40						
Calcium-44						

12. Write the electron configurations for the following ions and elements?

(4 points)

12.1 Bromide ion

12.2 Lithium ion

12.3 Potassium

12.4 Copper

13. Here is the hydrogen diagram Hydrogen



1s

Draw the complete orbital diagram for Mg, O²⁻, Na⁺

(3 points)

Mg

O²⁻Na⁺

Monrudee Phongaksorn

Parinya Khongprom

Sunanta Chuayprakong